

Greece Energy & Economy

Analysis Insights Report

Synthesized from energy, economy, DTW price,
and workforce notebooks

Generated June 22, 2026

Executive Summary

This report synthesizes findings from four Jupyter notebooks analyzing Greek energy markets, macroeconomic indicators, consumer price dynamics (2020–2023), and social/workforce trends.

Headline findings:

- Unemployment fell from a crisis peak of 27.7% (2013) to 11.0% (2023), signalling labour-market recovery.
- Wage indices (2020=100) moved from 114.1 in 2008 to 113.3 in 2024 (-0.7%), while CPI for many products rose far faster—creating a purchasing-power gap.
- Greece increasingly relies on imported energy; the share of traded energy in total consumption climbed through 2021–2023, aligning with broader price pressures.
- CPI increases were largely synchronous across product categories (2020–2023), with petroleum-derived goods and animal products moving in related patterns.
- Fertility declined from 1.42 (2008) to 1.26 (2024), despite post-2013 economic recovery—suggesting persistent social or cost-of-living constraints.

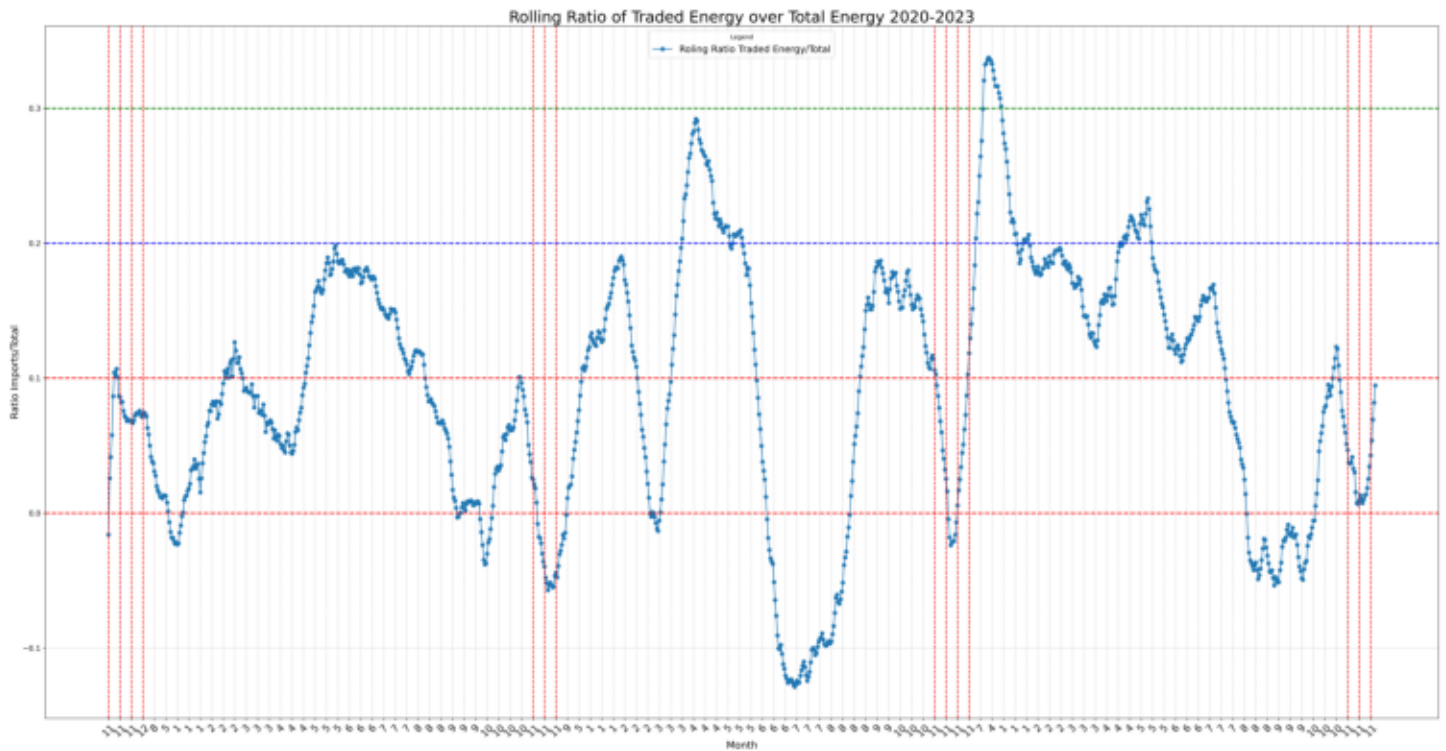
1. Energy Market Insights

Renewable generation (Nov 2020 – Jan 2023) averaged 1088 MWh per hour with peaks near 3113 MWh. Production follows strong seasonality: output drops in spring and rises in winter, visible in both rolling averages and violin plots of monthly distributions.

Domestic fuel mix shifted across lignite, gas, hydro, wind, and renewables, but net trade tells the stronger story. The ratio of imported to total energy is predominantly positive—Greece is a net buyer. By 2021, more than half the year exceeded 10% import dependence; in 2022 roughly half the year exceeded 10%; and in 2023 import share surpassed 30% for extended periods.

Implication: rising external energy dependence likely contributed to household and industrial cost pressures, reinforcing CPI increases documented in the price-analysis notebook.

Energy: Net trade vs total consumption

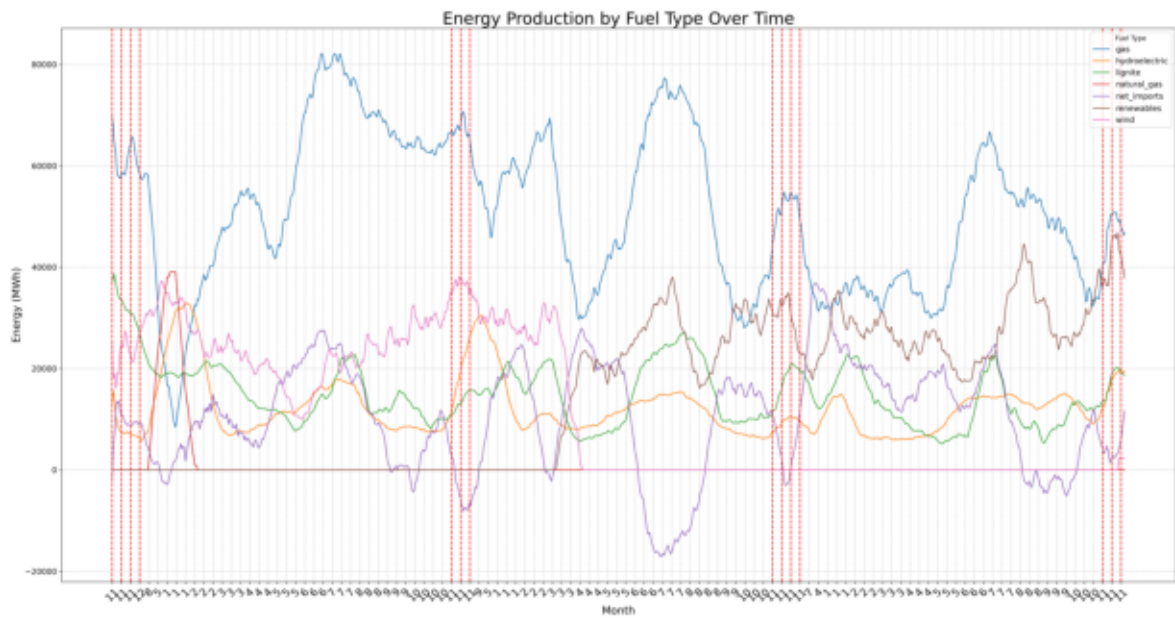


Key insight

Import share exceeded 30% for much of 2023—up from roughly 10–20% thresholds in prior years.

Ratio of traded (imported) energy to total energy by month. Positive values indicate net imports.

Energy: Source mix over time



Key insight

Seasonal winter peaks and spring troughs appear across sources; renewables grow but imports fill gaps.

Daily energy balance by fuel/source, showing seasonal and structural shifts across the study period.

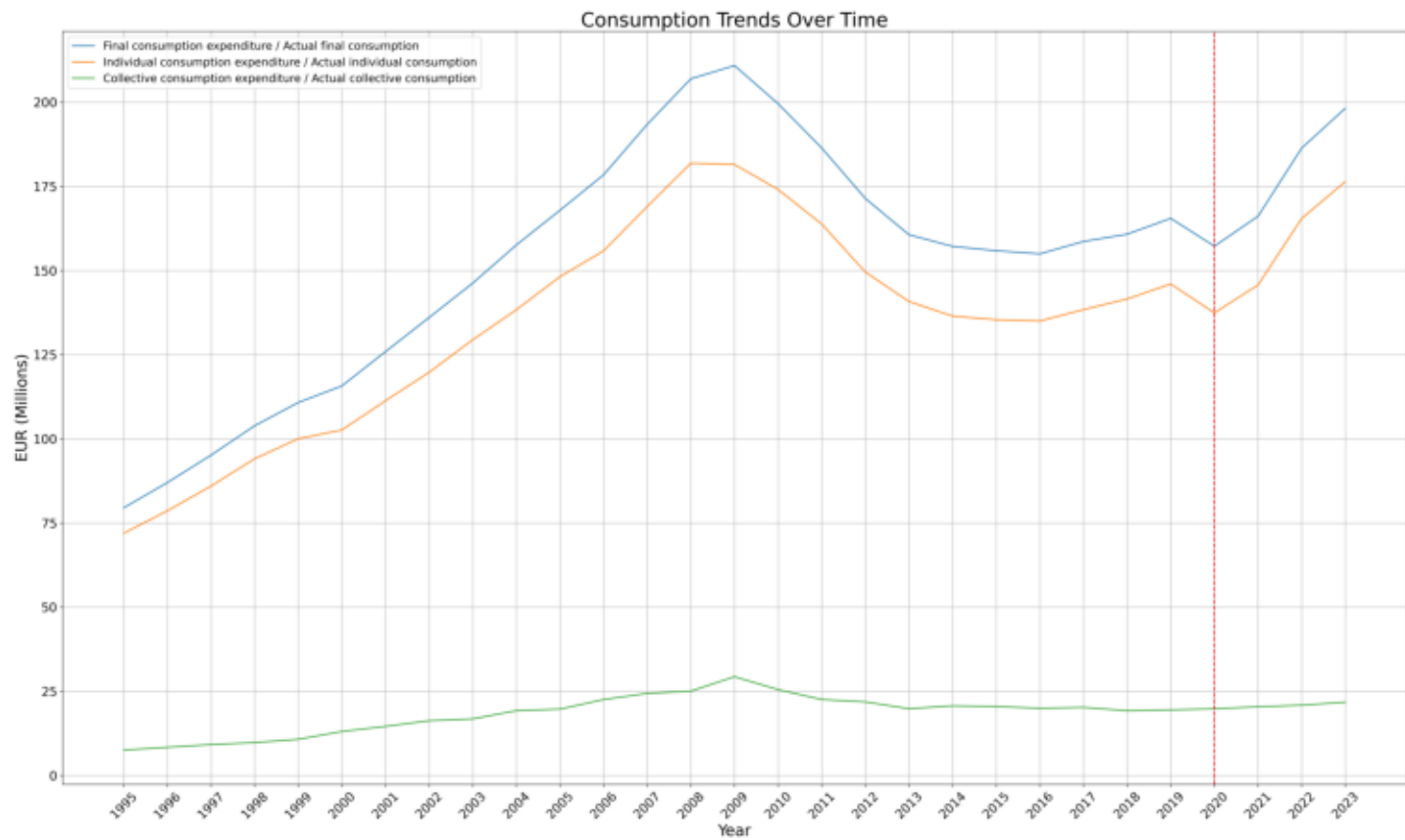
2. Macroeconomic Indicators

Sector-level national accounts (1995–2023) show Greece's long recovery arc. Production and employee compensation expanded substantially after the debt crisis, but the COVID-19 shock in 2020 produced a sharp dip in disposable income and private consumption.

Government collective consumption remained steadier than household spending during lockdowns—public services continued while discretionary consumption fell. Post-2020 data show recovery in consumption and investment, coupled with higher savings ratios as households adopted precautionary behaviour.

Debt dynamics (quarterly gross consolidated debt) remain a structural backdrop: recovery in activity did not eliminate legacy indebtedness, and fiscal balances reflect ongoing adjustment.

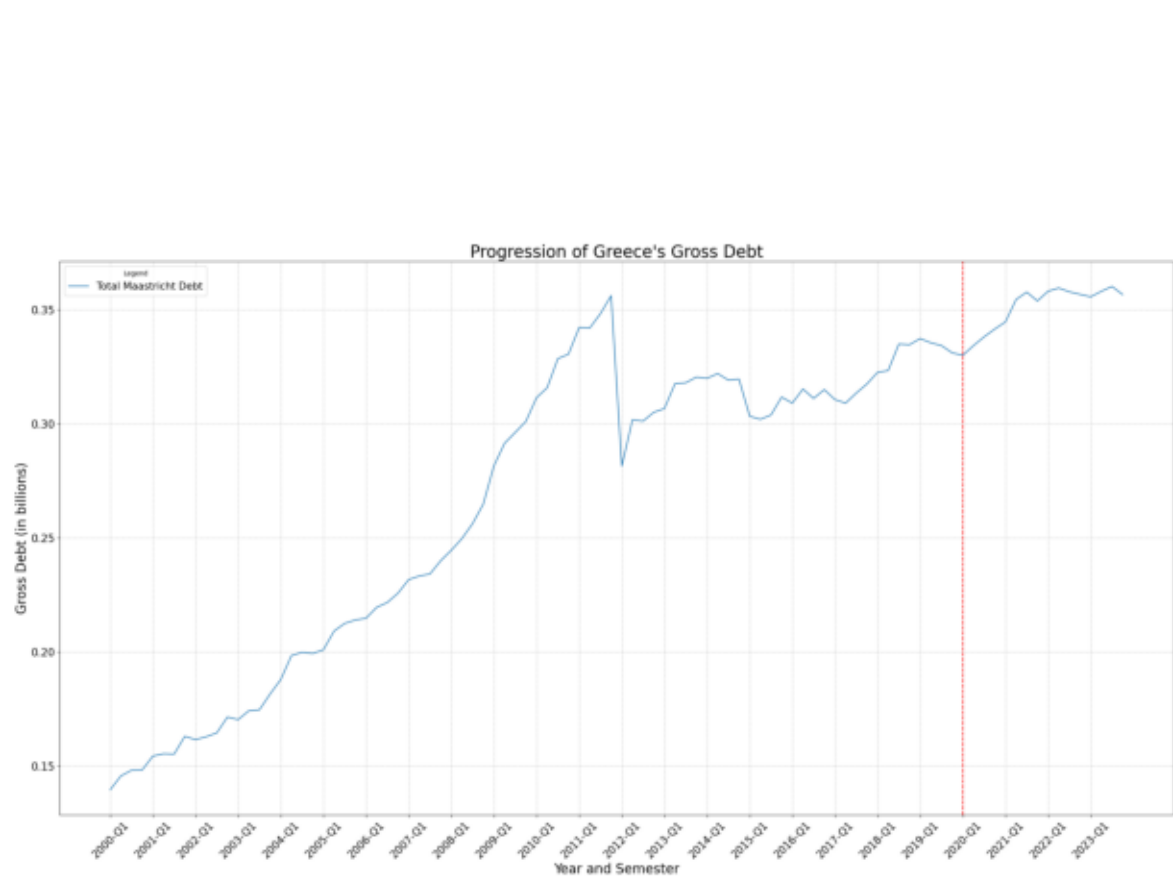
Economy: Consumption trends



Key insight

2020 lockdowns cut private consumption sharply; post-2020 lines show gradual normalization.

Economy: Debt progression



Key insight

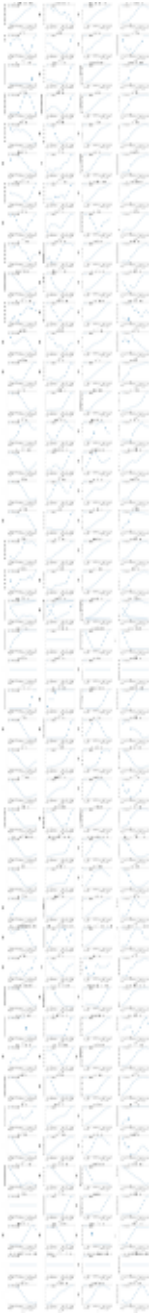
Debt remains elevated relative to pre-crisis levels—recovery is real but legacy burden persists.

3. Consumer Prices & Product Linkages (DTW)

Using harmonized CPI micro-data (2020–2023) for 45 essential products, analysis reveals:

1. Broad co-movement: Pearson correlation heatmaps show widespread positive co-movement—most product pairs rose together. This reflects economy-wide inflation rather than isolated category shocks. Correlation measures association, not causation.
2. Non-linear linkages: Distance correlation captures relationships Pearson misses. Petroleum products (heating oil, natural gas, solid fuels, gasoline) share market-driven patterns. Electricity shows more volatile, partially decoupled movement.
3. Granger causality: 289 product pairs show statistically significant lead-lag relationships ($p < 0.05$) at various lags—e.g. beef prices leading fresh produce and gasoline. These support forecasting models, not direct cause-effect claims.
4. Dynamic Time Warping (DTW): Low DTW distances highlight trend-aligned pairs—notably animal products with other edibles, and electricity with heating oil—useful for anticipating co-inflation episodes.

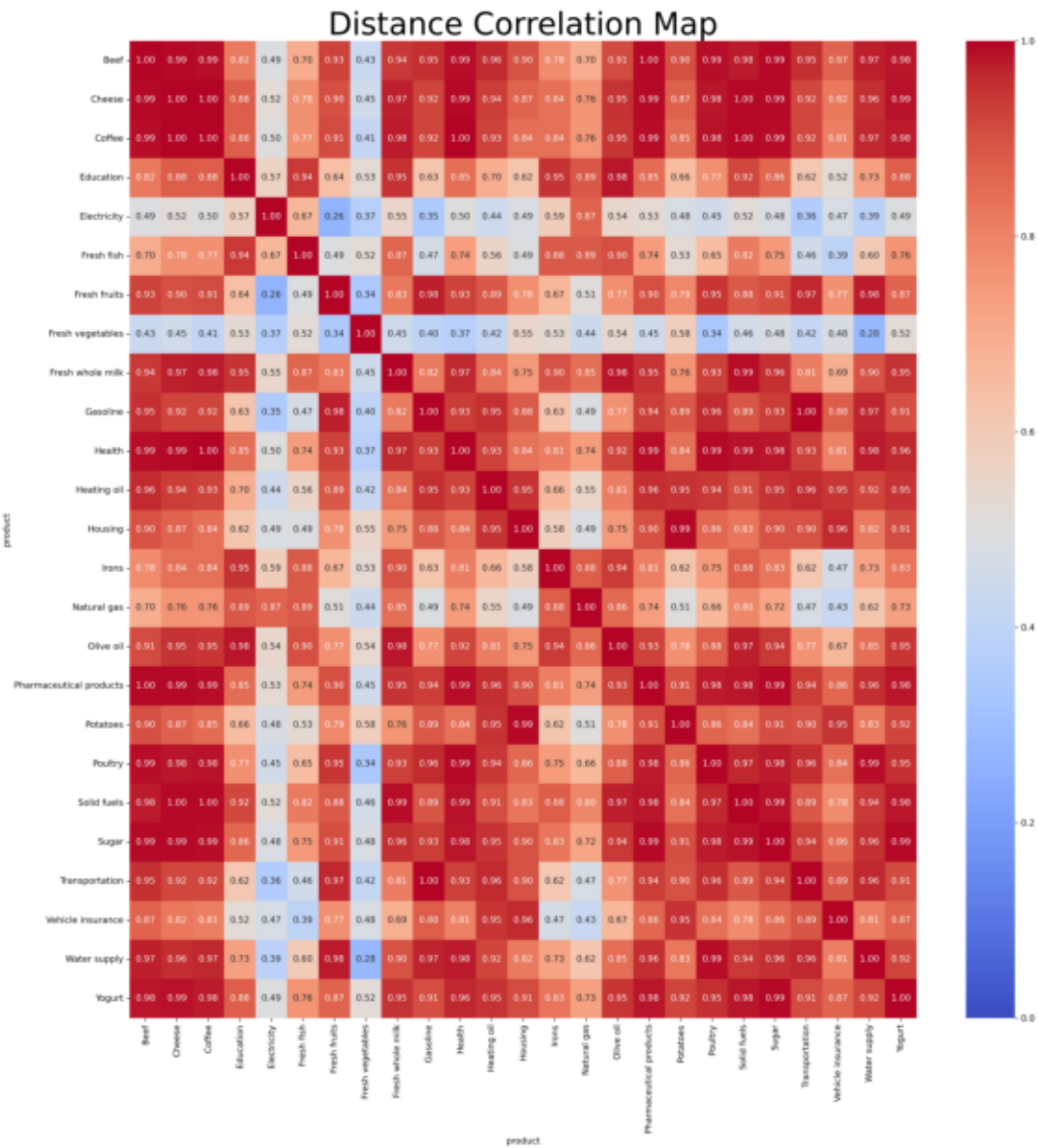
Prices: Year-over-year CPI paths



Key insight

Most product lines trend upward together—economy-wide inflation rather than isolated spikes.

Prices: Distance correlation structure



Key insight

Petroleum-related goods cluster; electricity shows distinct volatility patterns.

Distance-correlation heatmap capturing linear and non-linear co-movement between products.

4. Workforce, Wages & Social Indicators

Workforce restructuring: Employment in goods-producing sectors (agriculture, manufacturing) declined over decades, while services—especially seasonal and tourism-related work—grew. COVID-19 caused a sharp 2020 drop across sectors; all groups rebounded afterward, consistent with falling unemployment.

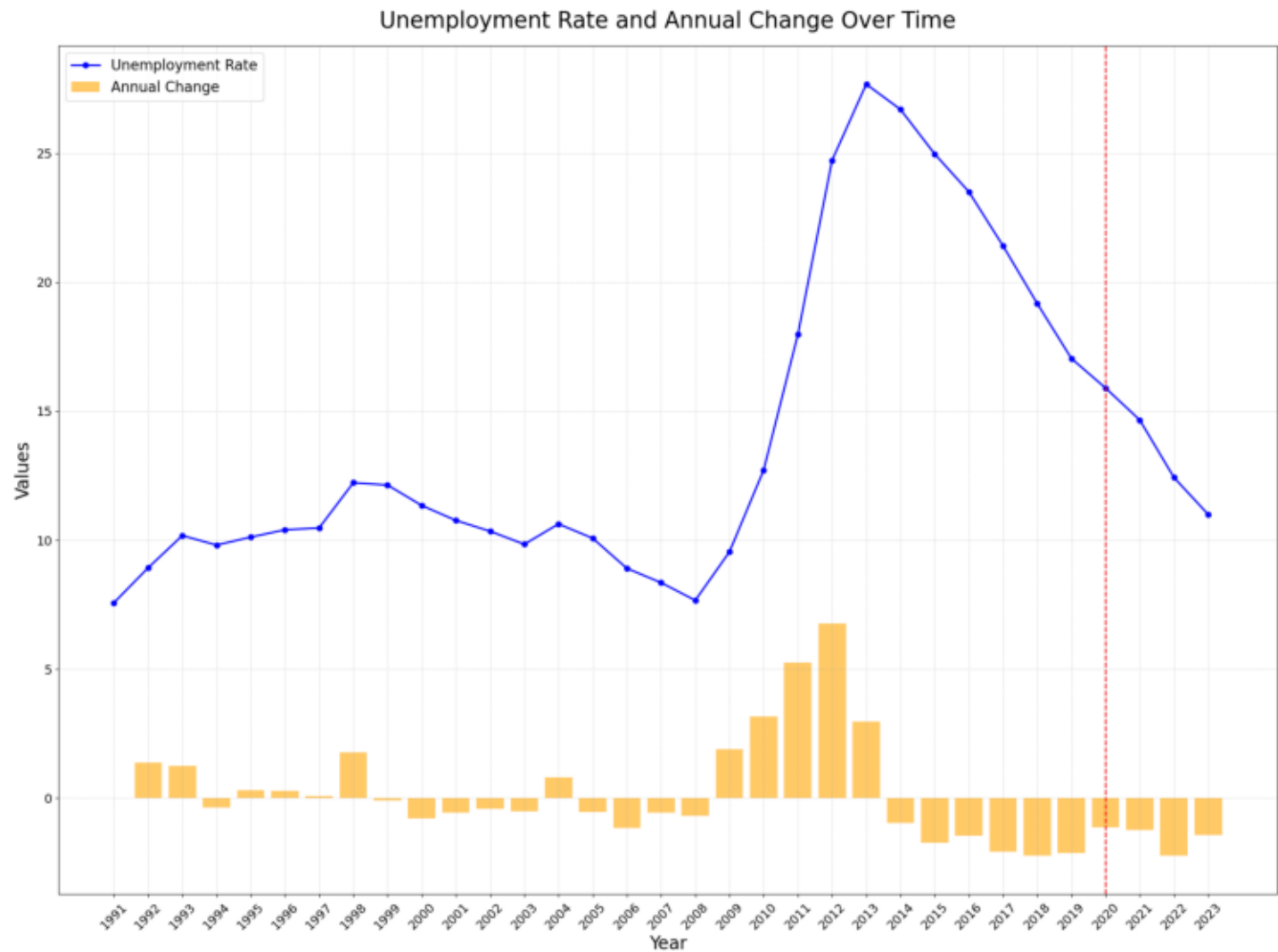
Unemployment peaked near 27% in 2013 (roughly one in four workers jobless). By 2023 the rate reached about 11%—a major improvement—even holding through the pandemic period.

Crime: Reported criminality rose gradually after COVID-19 despite improving labour metrics. This divergence may relate to cost-of-living pressure and shifts from domestic production toward imported consumption rather than unemployment alone.

Fertility: The rate fell ~ 0.25 since 2008 and continued declining through 2023 before stabilizing. Economic recovery alone did not restore birth rates—housing, childcare, and wage-price gaps may matter.

Wages vs prices: Minimum and average wage indices regressed during the 2008 crisis and only returned to pre-crisis levels by 2024, with a one-year COVID setback. Post-2020 wage gains ($\sim 15\%$) lagged CPI increases ($> 50\%$ for many products)—the central tension across this entire analysis.

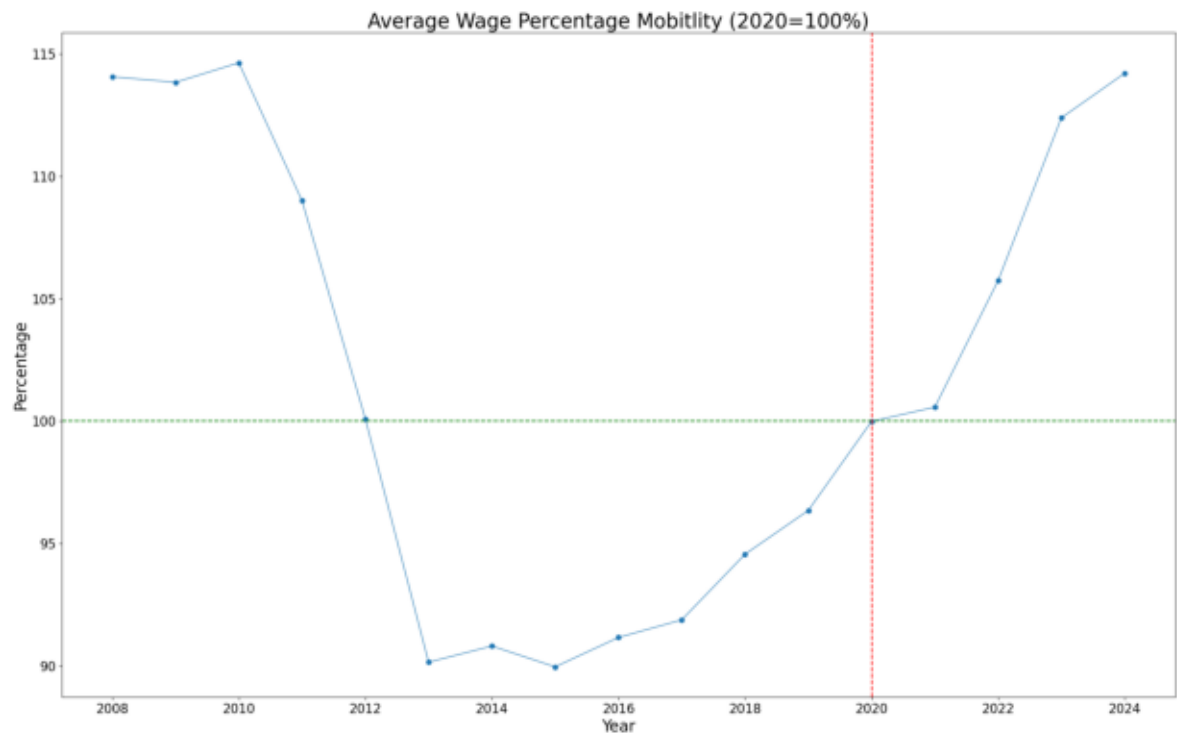
Labour: Unemployment trajectory



Key insight

Peak 27.7% (2013) → 11.0% (2023).

Wages: Index progression

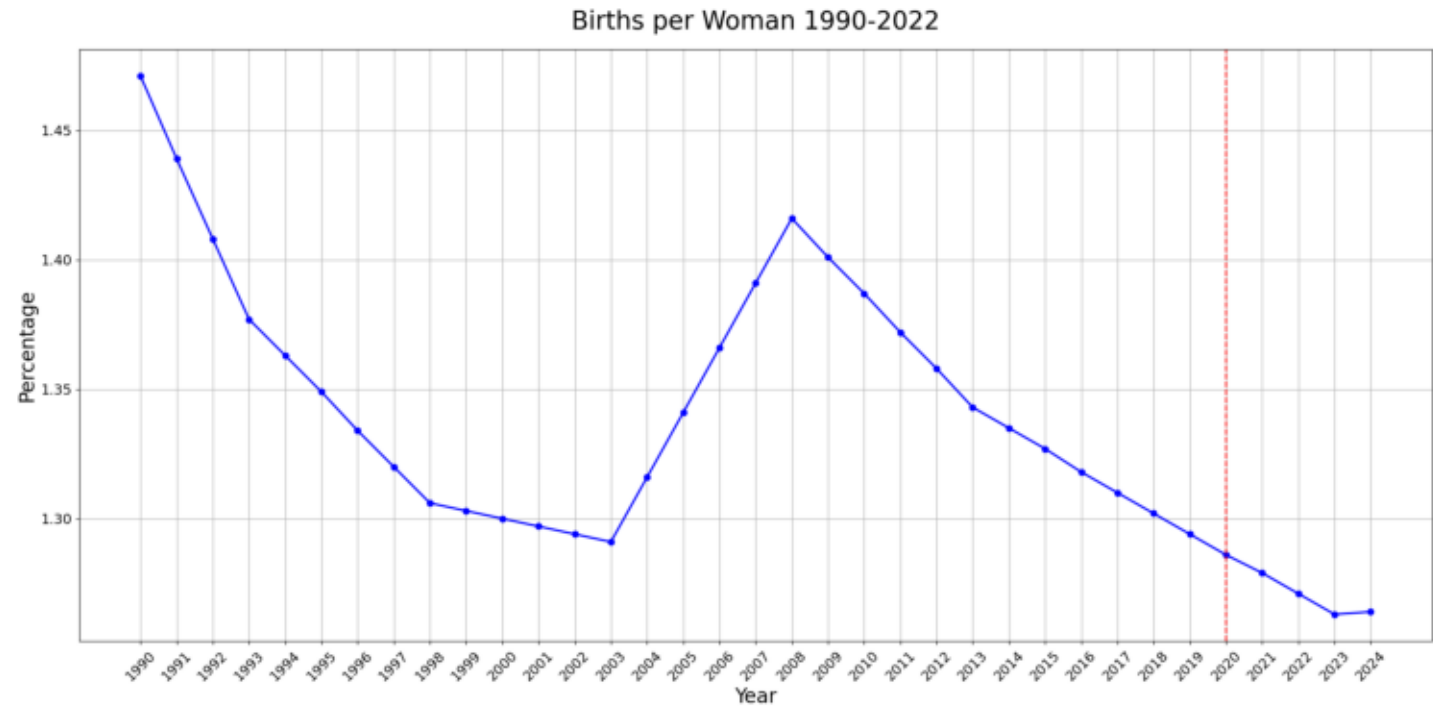


Key insight

Wage index ~114 (2008) vs ~113 (2024)—minimal net gain over 16 years.

Wage index progression (2020=100 base in source data).

Demographics: Fertility rate



Key insight

Decline from 1.42 (2008) to 1.26 (2024) despite labour-market recovery.

Conclusions & Policy Implications

Cross-cutting theme: headline macro indicators (GDP sectors, unemployment, production) point to recovery, but household-level stress persists through price inflation, energy import dependence, stagnant real wages, low fertility, and modest crime upticks.

Policy-relevant observations:

- Energy security and renewable expansion could reduce import-driven price pass-through.
- Wage indexation or targeted relief may be needed where CPI outpaces compensation.
- Labour shifted toward services/tourism while goods-producing employment shrank—structural, not cyclical.
- Demographic decline continues despite improved unemployment—monitor cost-of-living and family policy.

Data sources: ELSTAT/Eurostat-style CSVs in `data/`; notebooks in `notebooks/`; charts in `outputs/graphs/`. Generated automatically from notebook markdown conclusions and computed dataset statistics.